Somnath Bar, Ph.D.

Postdoctoral Scholar
Department of Civil and Environmental Engineering, University of California-Irvine, CA, USA
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Education

2018 – 2022 **Ph.D. in Geoinformatics, Central University of Jharkhand, India**
Thesis: *Assessment of Forest fire and its impact on regional atmospheric pollution and forest degradation in parts of Western Himalaya, India*
Key Projects: Fire emission quantification | Himalayan forest degradation assessment | Fire probability modeling | Fire-climate-human activity analysis | Small-isolated fire mapping (ML/GEE)

2015 – 2017 **Master's in Geography, Vidyasagar University, India**
Dissertations: Braided Teesta River channel dynamics using EO | Process-landform relationships at Ramkrishna Beach, Visakhapatnam

2012 – 2015 **Bachelor in Geography, Vidyasagar University, India**

Experience

Apr. 2024 – Present **Postdoctoral Scholar**, Department of Civil and Environmental Engineering, University of California-Irvine, US
Project: *Development of a fundamental research framework in wildfire through the lens of Earth Observation (EO), geospatial science, climate, and computational dynamics.*

Apr. 2024 – Sep. 2025 **Visiting Fellow**, School of Geography and Environmental Science, University of Southampton, UK

Jun. 2023 – Apr. 2024 **Research Fellow**, School of Geography and Environmental Science, University of Southampton, UK
Projects:

- *Development of country-scale vegetation products from Sentinel 2 in high spatial (100m) and temporal (15 days) resolution.*
- *Developing Ground-Based Observations for Validation (GBOV) vegetation products (fAPAR, LAI, fCover) for Copernicus Global Land Products.*
- *Earth observation data in monitoring orangutan habitat in Sumatra, Indonesia.*

Dec. 2022 – Apr. 2023 **Associate (Climate Data Scientist and Analyst)**, Climate Resilient Hub, WTW, Mumbai
Project: *Wildfire risk prediction model using physical and statistical methods.*

Research Interests

• Land-atmosphere interaction, • Climate Change • Geo-spatial Science • Remote Sensing • Sustainability • Geography, • Wildfire dynamics, • Geo-intelligence

Awards, Honors & Fellowships

Jun. 2022 — **Best Paper Award**, CUJ-ISG International Conference 2022
"Altering forest fire regime due to changing climatic pattern over south and central Indian Agro-Climatic regions" at the International Conference on Geospatial Pathways and Big Data Analytics in Natural Resource Applications and Climate Change.

Dec. 2017 — **National Eligibility Test (NET)** with **Junior Research Fellowship (JRF)** in Earth Science,
Council of Scientific & Industrial Research (CSIR-India)

Jan. 2017 — **National Eligibility Test (NET)** with **Junior Research Fellowship (JRF)** in Geography,
University Grants Commission (UGC-India)

Skills

Programming & Scripting	Python, R, Google Earth Engine (GEE), ArcPy, GrADS, LaTeX
Geospatial Platforms	QGIS, SNAP, GEE, ArcGIS, ENVI, SAGA GIS
Data Sources & Formats	MODIS, VIIRS, Sentinel-2, Landsat, UAV, LiDAR, SAR; MERRA-2, ERA5; NetCDF, HDF
Modeling & Simulation	WRF, WRF-SFIRE, FlamMap, MaxEnt
Scientific Writing	LaTeX, manuscript preparation, peer-reviewed publishing,
Additional Skills	Linux, ML and DL, Unmanned Aerial Vehicle (UAV), team collaboration, and academic mentoring

Research Publications (*Citations: 1012+, h-index: 14*)

Journal Articles

1. **S. Bar**, S. Li, and T. Banerjee, "Explaining building damage from wildfires in california", *Science Advances [In-Press]*, 2026. doi:10.1126/sciadv.aed4197 (, IF: 13.9 , Q1).

2. M. Rivera, L. E. Lowman, A. Mrad, **S. Bar**, and T. Banerjee, "Characterizing wildfire behavior with ecostress land surface temperature across four california case studies", *Ecological Informatics*, 2026. URL: `https://doi.org/10.1016/j.ecoinf.2026.103777` (, IF: 8.5 , Q1).

3. B. Paul, U. C. Dumka, and **S. Bar**, "Forest fire regimes in the Northwestern Himalayas: Unravelling microlevel impact of topography, weather, and human activity on fire behaviour", *International Journal of Remote Sensing*, 2025. doi:10.1080/01431161.2025.2492413 (, IF: 2.8 , Q2).

4. **S. Bar**, P. Acharya, B. R. Parida, *et al.*, "Investigation of fire regime dynamics and modeling of burn area over India for the twenty-first century", *Environmental Science and Pollution Research*, 2024. doi:10.1007/s11356-024-32922-w (, IF: 4.463 , Q1).

5. B. Kanu, B. R. Parida, **S. Bar**, C. S. Dwivedi, and A. C. Pandey, "Estimating forest biophysical and biochemical parameters in Behali Reserve Forest (Assam) using proximal and remote sensing techniques", *Tropical Ecology*, 2024. doi:10.1007/s42965-024-00359-4 (, IF: 1.9 , Q2).

6. T. Mahato, B. R. Parida, and **S. Bar**, "Assessing tea plantations biophysical characteristics in Northeast India using satellite data", *Environmental Monitoring and Assessment*, 2024. doi:10.1007/s10661-024-12502-8 (, IF: 3.6 , Q2).

7. A. Maiti, M. K. Hasan, S. Sannigrahi, *et al.*, "Optimal rainfall threshold for monsoon rice production in India varies across space and time", *Communications Earth & Environment*, 2024. doi:10.1038/s43247-024-01414-7 (, IF: 9.3 , Q1).

8. **S. Bar**, B. R. Parida, A. C. Pandey, *et al.*, "Modeling and prediction of fire occurrences along an elevational gradient in Western Himalayas", *Applied Geography*, 2023. doi:10.1016/j.apgeog.2022.102867 (, IF: 7.4 , Q1).

9. **S. Bar**, B. R. Parida, A. C. Pandey, and N. Kumar, "Pixel-based long-term (2001–2020) estimations of forest fire emissions over the Himalaya", *Remote Sensing*, 2022. doi:10.3390/rs14215302 (, IF: 4.3 , Q1).

10. A. Maiti, P. Acharya, S. Sannigrahi, *et al.*, "Mapping active paddy rice area over monsoon Asia using Sentinel-2 images", *Geocarto International*, 2022. doi:10.1080/10106049.2022.2032396 (, IF: 3.9 , Q2).
11. S. Sannigrahi, F. Pilla, A. Maiti, *et al.*, "Examining the status of forest fire emission in 2020 and its connection to COVID-19 incidents in West Coast regions of the United States", *Environmental Research*, 2022. doi:10.1016/j.envres.2022.112818 (, IF: 8.2 , Q1).
12. P. Acharya, G. Barik, B. K. Gayen, *et al.*, "Revisiting the levels of aerosol optical depth in south-southeast Asia, Europe and USA amid the COVID-19 pandemic", *Environmental Research*, 2021. doi:10.1016/j.envres.2020.110514 (, IF: 8.2 , Q1).
13. **S. Bar**, B. R. Parida, S. P. Mandal, A. C. Pandey, N. Kumar, and B. Mishra, "Impacts of partial to complete COVID-19 lockdown on NO2 and PM2.5 levels in major urban cities of Europe and USA", *Cities*, 2021. doi:10.1016/j.cities.2021.103308 (, IF: 7.0 , Q1).
14. **S. Bar**, B. R. Parida, G. Roberts, A. C. Pandey, P. Acharya, and J. Dash, "Spatio-temporal characterization of landscape fire in relation to anthropogenic activity over the Western Himalaya, India", *GIScience & Remote Sensing*, 2021. doi:10.1080/15481603.2021.1879495 (, IF: 6.9 , Q1).
15. B. R. Parida, **S. Bar**, D. Kaskaoutis, A. C. Pandey, S. D. Polade, and S. Goswami, "Impact of COVID-19 induced lockdown on land surface temperature in Europe and North America", *Sustainable Cities and Society*, 2021. doi:10.1016/j.scs.2021.103336 (, IF: 13.3 , Q1).
16. B. R. Parida, **S. Bar**, G. Roberts, *et al.*, "Improvement in air quality during the first lockdown of the pandemic", *Environmental Research*, 2021. doi:10.1016/j.envres.2021.111280 (, IF: 8.2 , Q1).
17. B. R. Parida, **S. Bar**, N. Singh, B. Oinam, A. C. Pandey, and M. Kumar, "A short-term decline in anthropogenic emission of CO2 in India due to COVID-19 confinement", *Progress in Physical Geography: Earth and Environment*, 2021. doi:10.1177/0309133320966741 (, IF: 3.5 , Q2).
18. S. Thakur, I. Mondal, **S. Bar**, *et al.*, "Shoreline changes and its impact on the mangrove ecosystems of Indian Sundarbans", *Journal of Cleaner Production*, 2021. doi:10.1016/j.jclepro.2020.124764 (, IF: 10.7 , Q1).
19. **S. Bar**, B. R. Parida, and A. C. Pandey, "Forest fire burn area mapping using machine learning algorithms on GEE cloud platform over Uttarakhand, Western Himalaya", *Remote Sensing Applications: Society and Environment*, 2020. doi:10.1016/j.rsase.2020.100324 (, IF: 4.9 , Q1).
20. G. Barik, P. Acharya, A. Maiti, B. K. Gayen, **S. Bar**, and A. Sarkar, "A synergy of linear model and wavelet analysis of aerosol optical depth during pre-monsoon season over Indian sub-continent", *Journal of Atmospheric and Solar-Terrestrial Physics*, 2020. doi:10.1016/j.jastp.2020.105478 (, IF: 2.2 , Q2).

Preprints

1. T. Mahato, S. Sarkar, B. R. Parida, and **S. Bar**, "Beyond multispectral limits: Satellite embeddings for fire severity estimation across global biomes", Preprint, 2026. doi:10.21203/rs.3.rs-9849270/v1.

Conference Proceedings

1. C. Lerebourg, R. Grousset, T. Vidal, *et al.*, "GBOV service: Latest product updates for EO data validation", in *EGU General Assembly 2024*, 2024. doi:10.5194/egusphere-egu24-10447.
2. **S. Bar**, B. R. Parida, and B. U. Shankar, "Unfolding the contribution of environmental variables in forest fire over western Himalaya", in *IEEE International India Geoscience and Remote Sensing Symposium*, 2021. doi:10.1109/InGARSS51564.2021.9792002.

Book Chapters

1. **S. Bar**, B. R. Parida, and J. Dash, "Disentangle the short-term forest degradation over most fire-affected parts of western himalaya, india", in *Handbook of Himalayan Ecosystems and Sustainability, Volume 1*. CRC Press, 2022. doi:10.1201/9781003268383-7.

2. **S. Bar**, B. R. Parida, A. C. Pandey, and S. K. Panda, “Changing forest fire regime in relation to climatic conditions over western and eastern himalaya, india”, in *Handbook of Himalayan Ecosystems and Sustainability, Volume 1*. CRC Press, 2022. doi:10.1201/9781003268383-19.
3. B. R. Parida, S. Sparsha, **S. Bar**, A. C. Pandey, and N. Kumar, “Monitoring land use/land cover change and high-altitude vegetation trends along with their climatic controls across the central and eastern himalayas”, in *Handbook of Himalayan Ecosystems and Sustainability, Volume 1*. CRC Press, 2022. doi:10.1201/9781003268383-15.
4. **S. Bar**, B. Kumari, and S. K. Gupta, “Salinization of coastal groundwater resource in the perspective of climate change”, in *Climate Change and Coastal Ecosystems*. Springer Singapore, 2021. doi:10.1007/978-981-15-6564-9_17.

Manuscripts Under Preparation

1. **S. Bar**, “Does a damaged building increase its neighbors’ wildfire damage risk? evidence from the 2025 palisades and eaton fires”, In Preparation, 2026.
2. **S. Bar**, “Relative presence of fine and woody fuel: Influence on fire behavior and fire regime over conus”, In Preparation, 2026.
3. M. Rivera, **S. Bar**, M. Pascolini-Campbell, *et al.*, “Post-wildfire ecosystem resilience in prescribed burn regions revealed by ecostress et”, In Preparation, 2026.

Conference/Workshop Presentations

2026 — **Explaining Building Damage from Wildfire across California**, 10th Annual Postdoctoral Scholar Research Symposium, University of California-Irvine, April 2026.

2025 — **Key Predictors of Wildfire Building Damage - A Data-Driven Approach Across California’s WUI**, American Geophysical Union (AGU), New Orleans, USA, Dec 2025.

- **Fire Ecology from Field to Drone: Hayfork Campaign**, IFireNET-2025, Department of Civil and Environmental Engineering, University of California-Irvine (UCI), USA.
- **Influence of Vegetation Biophysical Properties and Plant Functional Type Composition on Wildfire**, IEEE IGARSS, Brisbane 2025.
- **Building Flammability and Atmospheric Dryness as Dominant Predictors of Wildfire Building Damage: A Data Driven Approach Across California’s WUI**, HydroML, Lake Arrowhead 2025.

2024 — **Characterizing the role of plant functional types in Influencing wildfire behavior across the United States**, American Geophysical Union (AGU), Washington DC, USA, Dec 2024.

2023 — **Impact of climatic drought of 2022 on vegetation biophysical traits across the UK**, RSPSoc Conference, British Geological Survey, UK.

2022 — **Altering forest fire regime due to changing climatic pattern**, CUJ-ISG International Conference on Geo-spatial Pathways, India.

2021 — **Unfolding environmental and anthropogenic variables in forest fire**, IEEE InGARSS, India.

2020 — **Synergetic relation of Forest fire and vegetation traits**, AGU Fall Meeting, USA.

- **Forest fire and atmospheric pollutants over Western Himalaya**, ISRO-SAC Symposium, Ahmedabad, India.
- **Variabilities of Western Himalayan Forest Fire**, JpGU-AGU Joint Meeting, Japan.

2019 — **Unsupervised and supervised classification for forest fire patches**, AGU Fall Meeting, San Francisco, USA.

- **Forest fire identification using machine learning on GEE**, NESAC Symposium, Shillong, India.

Invited Talks/Lectures

2025 — **Concepts of Fire Ecology: Understanding and Managing Indian Forest Fires in a Changing Climate**, Belda College, Vidyasagar University, India; 30 June 2025.

- **Indian forest fire dynamics & fuel management**, ARIES, India, 27 Feb 2025.
- **EO for Vegetation ECVs to Wildfire**, Central University of Jharkhand, India, 18 Feb 2025.

2024 — **Global forest fire mechanisms & Indian perspective**, Rashtriya Raksha University, India, 25 July 2024.

- **Connecting Dots in Geography: Insight from EO**, Vidyasagar University, India, 19 July 2024.
- **Forest Fire Mechanisms & Management**, Disaster Risk Reduction Course, Rashtriya Raksha University, India, March 2024.

2022 — **Tracing Forest Fire Clusters using GEE**, Central University of Jharkhand, India, June 2022.

Professional Service

Reviewer — • Agricultural and Forest Meteorology • Building and Environment • Current Science • Ecological Indicators • Estuaries and Coasts • Remote Sensing Applications: Society and Environment • Remote Sensing Letters • Forest • Tropical Ecology • Trees, Forests and People • Journal of Cleaner Production

Conference Organization — **Session Organizer, IEEE IGARSS 2026** — *"Wildfire and Earth Observation: Remote Sensing for Ecosystem Resilience and Hazard Adaptation with Artificial Intelligence (AI)"*, Washington, DC, USA, August 2026

Research Impact

Media Coverage and Outreach — Quoted as an expert in *Down To Earth* (March 2026) article, "Himalayan fire shift: Tale of flames and ice unfolding across the range", based on an interview highlighting satellite-based evidence of increasing high-elevation forest fires (2,000–4,000 m) across the Himalayas.

Professional Development

2026 — **Mentoring Excellence Program**, 5-week certificate program, **University of California, Irvine Graduate Division**; 16 Apr–14 May.

2019 — **Data Analysis using AI and Machine Learning**, ISG Pre-symposium Tutorial, NESAC, Shillong, Meghalaya, India; 18–19 Nov

- **Global Climatic Anomalies and Teleconnections & Regional Implication**, TERI, New Delhi, India; 5–8 Aug, (International Climate Research School, Research Council of Norway & MoES India)
- **Earth Observation Data for Forest and Agriculture Carbon Modelling**, CORAL, IIT Kharagpur, India; 18–22 Feb, (International Course)

2018 — **Research Methodology and LaTeX Learning**, Dept. of Computer Science and Technology, Central University of Jharkhand, India (One-day Workshop)

- **Qualitative & Quantitative Research Methods in Geographical Studies**, Dept. of Geography, Vivekananda College, Thakurpukur, Kolkata (One-day Workshop)
- **Advanced Numerical Modelling of Meteorology and Emissions for Air Quality Modelling**, JNTU College, Telangana, India; 29 Oct–2 Nov 2018 (GIAN, MHRD, Govt. of India)
- **Numerical Weather Prediction**, NIT Warangal, Telangana, India; 18–29 June 2018 (GIAN, MHRD, Govt. of India)

2017 — **Air Pollution Control Techniques: Design and Development**, NIT Jalandhar, Punjab, India; 11–15 Sep 2017 (GIAN, MHRD, Govt. of India)

2016 — **Active Continental Tectonics**, IISER Kolkata, West Bengal, India; 6–19 Nov 2016 (GIAN, MHRD, Govt. of India)